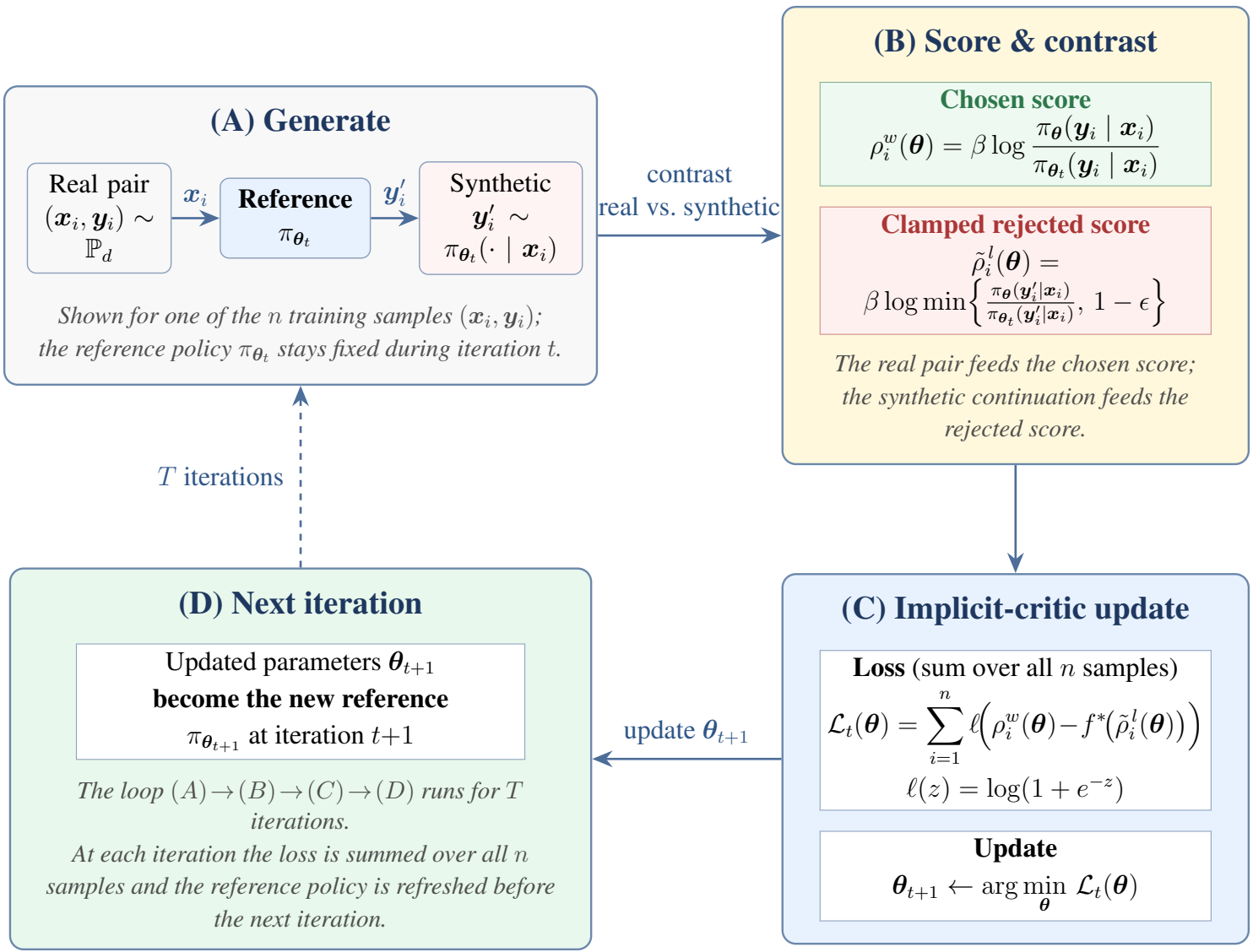

(A) Generate
Real pair $(x_i, y_i) \sim \mathbb{P}_d$
x_i
Reference π_{θ_t}
y'_i
Synthetic $y'_i \sim \pi_{\theta_t}(\cdot \mid x_i)$
Shown for one of the n training samples (x_i, y_i); the reference policy π_{θ_t} stays fixed during iteration t.
contrast real vs. synthetic
(B) Score & contrast
Chosen score
$\rho_i^w(\theta) = \beta \log \frac{\pi_\theta(y_i \mid x_i)}{\pi_{\theta_t}(y_i \mid x_i)}$
Clamped rejected score
$\tilde{\rho}_i^l(\theta) = \beta \log \min\left\{\frac{\pi_\theta(y'_i|x_i)}{\pi_{\theta_t}(y'_i|x_i)}, 1-\epsilon\right\}$
The real pair feeds the chosen score; the synthetic continuation feeds the rejected score.
(C) Implicit-critic update
Loss (sum over all n samples)
$\mathcal{L}_t(\theta) = \sum_{i=1}^{n} \ell\Big(\rho_i^w(\theta) - f^*\big(\tilde{\rho}_i^l(\theta)\big)\Big)$
$\ell(z) = \log(1 + e^{-z})$
Update
$\theta_{t+1} \leftarrow \arg\min_\theta \mathcal{L}_t(\theta)$
update θ_{t+1}
(D) Next iteration
Updated parameters θ_{t+1} become the new reference $\pi_{\theta_{t+1}}$ at iteration $t+1$
The loop $(A) \to (B) \to (C) \to (D)$ runs for T iterations.
At each iteration the loss is summed over all n samples and the reference policy is refreshed before the next iteration.
T iterations